

Longitudinal Genomic Characterization of Carbapenemase-producing Enterobacteriaceae (CPE) Reveals Changing Pattern of CPE Isolated in Hong Kong Hospitals

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ABSTRACT

An active, territory-wide, CPE surveillance program implemented from 2011 showed increasing levels of carbapenemase-producing Enterobacteriaceae (CPE) isolates from patients in Hong Kong hospitals. The molecular epidemiology of 567 CPE from patients of three of seven public hospital clusters in Hong Kong are described. During a 7-year period, the incidence of CPE isolation increased from 0.05 to 9.6/100 000 patient-days. The carbapenemase genes identified were polyclonal, including *bla*_{KPC}, *bla*_{NDM} and *bla*_{IMP}, which were mainly associated with hospitalization overseas in previous years. However, increasing CPE isolation from patients without hospitalization overseas occurred in 2015, with *bla*_{NDM} (52.6%) predominant followed by *bla*_{IMP} (30.0%). *Escherichia coli* (46.4%) and *Klebsiella* spp. (38.3%) were the dominant species. Whole-genome sequencing was performed on 169 representative isolates with a combination of short and long reads using Illumina and Nanopore technology. Two distinct lineages of *bla*_{KPC-2}-positive *Klebsiella pneumoniae* (ST11 and ST258) were identified with ST11 carrying yersiniabactin gene *ybt-9* on ICEKp3. ST131 *E. coli* producing IMP-4 was present throughout the study period. The *bla*_{NDM} and *bla*_{IMP} genes were mainly carried in IncX3 and IncN-ST7 plasmids, respectively. *bla*_{OXA-48-like} gene was carried in the IncX3 plasmid in *E. coli* and in the ColKP3 plasmid in *K. pneumoniae*. A lineage of *K. pneumoniae* with *bla*_{NDM-1} plus *bla*_{OXA-232} in distinct plasmids of IncF1B/IncHI1B was identified and associated with prior hospitalization overseas. This study highlights the threat of multiple types of CPE, with the predominance of *bla*_{NDM} and *bla*_{IMP} among CPE in our hospitals. Enhanced containment strategies are needed to mitigate the trend of rapidly rising CPE in healthcare settings.

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1. Introduction

Carbapenemase-producing Enterobacteriaceae (CPE) are a pressing and increasing global public health concern. Horizontal gene transfer of mobile genetic elements carrying these genes can occur in CPE and infections with these strains are associated with poor clinical outcome [1,2]. Although new antimicrobial agents are in accelerated development, understanding the molecular characteristics of CPE and the carbapenemases involved is crucial for implementing appropriate infection control and treatment intervention.

CPE were first detected in Hong Kong in 2009 when the Public Health Laboratory confirmed four isolates carrying genes encoding

carbapenemases among carbapenem-non-susceptible strains. The four isolates were an *bla*_{NDM-1} in *Escherichia coli*, *bla*_{IMI-3} in *Enterobacter cloacae*, and *bla*_{IMP-4} in *Klebsiella pneumoniae* and *Citrobacter freundii* [3]. To increase monitoring of CPE, an active surveillance program was implemented across public hospitals to screen for CPE carriage in at-risk patients according to CDC guidance. A previous study targeting carbapenem-resistant Enterobacteriaceae (CRE) from patients who had been hospitalized outside Hong Kong within the past 12 months or who had received surgery abroad revealed a fecal carriage rate of 1.2% [4].

Carbapenemase genes are frequently encoded on small transposable elements that can mobilize among plasmids, causing rapid non-clonal dissemination [5,6]. *bla*_{KPC} is the most prevalent class A carbapenemase gene and is usually carried by transposon Tn4401 with the ability to form recombinants with Tn2 transposon in western countries, whereas Tn1721 is frequently reported in China [5,7,8]. *bla*_{NDM} and *bla*_{IMP} belong to class B beta-lactamases and are mobilized within the insertion sequence IS*Aba125* and IS26 [9–11]. *Bla*_{OXA-48-like} group is another pandemic enzyme that is derived from the chromosome of *Shewanella* spp. and can transpose to plasmids through Tn1999 or Tn2013 [12]. As transposons generally consist of repeat regions, whole-genome sequence (WGS) using a combination of short-read and long-read sequencing enables the reconstruction of mobile genetic elements commonly associated with these carbapenemase genes.

The decreasing cost of next generation sequencing and ease of bioinformatics tools facilitate the Pan-genome analysis of large CPE surveillance studies [13,14]. In Europe, a genomic analysis of over 1700 carbapenem-resistant *K. pneumoniae* revealed the epidemic was due to nosocomial spread rather than inter-country transmission [15]. Another WGS study reported *K. pneumoniae* ST307 harboring *bla*_{OXA-181}-positive IncX3 plasmid caused nosocomial outbreaks in South African hospitals [6]. Furthermore, Nanopore long-read sequencing enables the generation of complete plasmids by hybrid assembly with short-read sequences, as demonstrated by a surveillance study on *K. pneumoniae* from South and Southeast Asia [16,17].

Surveillance reports from other Asian countries show the emergence of CPE but often lack comprehensive genomic analyses concerning plasmids and the profiles of antimicrobial resistance (AMR) genes or virulence factors (VFs) [14,18–20]. Here we report the epidemiology of CPE isolated from three of seven hospital clusters of Hong Kong over a 7-year period. Representative CPE strains were analysed using either Illumina short reads alone or in combination with long reads from MinION to characterize the molecular types and AMR genes and reconstruct the plasmids by hybrid assembly.

2. Material and Methods

2.1. Description of bacterial isolates

From 2011 to 2017, a total of 567 non-duplicate CPE isolates from patients hospitalized in seven hospitals from three of seven clusters of hospitals under the Hospital Authority (HA) were included. HA manages 43 public hospitals in seven clusters, and the three clusters in this study cover a catchment area of approximately half of Hong Kong's 7.5 million population. The active, territory-wide, surveillance program began in 2011. Essentially, at-risk patients who were admitted to the hospitals had a rectal swab obtained at admission for detection of CPE carriage [4]. If the patient screened positive, contact tracing was performed, and patients were isolated with control measures. At-risk patients included those with a history of hospitalization outside Hong Kong during the past 6 months (and extended to 12 months from 2016 according to the recommendation of Centers for Disease Control and Prevention [<http://www.cdc.gov/hai/pdfs/>

[cre/CRE-guidance-508.pdf](http://www.cdc.gov/hai/pdfs/cre/CRE-guidance-508.pdf)]) or contact case of CPE on re-admission. Additional sentinel screening of stool specimens submitted for *Clostridium difficile* testing, patients with hospitalization for over 14 days, or those hospitalized at another hospital cluster in the previous 3 months were added in 2016, and surgical admission screen was added in 2017. Since 2015, ChromID CARBA/OXA screening media generally replaced the conventional selective media. Suspected CPE isolates were routinely sent to the Public Health Laboratory Centre (PHLC) of the Centre for Health Protection, Department of Health, HKSAR for detection of carbapenemase. In-house developed polymerase chain reaction (PCR) was conducted by PHLC to detect both serine carbapenemase and metallo-beta-lactamases (MBLs) based on previous methods [21,22]. Ethics approval was obtained from the Joint CUHK-NTEC Ethics Committee and the respective Cluster Ethics Committees of Kowloon Central and Kowloon East Hospitals (CREC No.: KC/KE-17-0203/ER4).

2.2. Whole-genome sequencing and de novo assembly

WGS was performed for 169 strains. These included all strains from clinical samples and all carriage strains that detected more than one carbapenemase gene type. In addition, one strain of each of the four major carbapenemases (NDM, IMP, KPC and OXA-48-like) from each year per hospital cluster was included for WGS. Bacterial DNA was extracted using Wizard® genomic DNA purification kit (Promega, Madison, WI, USA) after an overnight culture of strains on blood agar at 37°C. DNA library preparation was performed with either Illumina Nextera XT DNA library preparation kit (Illumina, USA) or Riptide™ High throughput rapid library prep kit (iGenomx, USA) according to the manufacturers' instructions prior to sequencing on Illumina HiSeq or Miseq platform. DNA reads were analysed as previously described and with modifications as indicated [23]. Low-quality reads and adapters were trimmed by TrimGalore (v0.4.5, <https://github.com/FelixKrueger/TrimGalore>) and assembled by SPAdes (v3.12.0) using default parameters [24]. QUAST (v5.0.2) was used to summarize the assembly results [25]. Taxonomic classification was performed by both GTDBTk (v0.3.2) and Kraken2 (v2.0.8) [26,27].

2.3. Profiling of AMR genes and VFs

NCBI AMRFinder database and VFDB were used to detect antimicrobial resistance genes (ARGs) and virulence factors using ABRicate (v0.9.8, available at: <https://github.com/tseemann/abricate>) [28]. PlasmidFinder and pMLST were used for plasmid replicon detection and typing [29]. Genes with less than 95% coverage or 98% identity were excluded to minimize false-positive calls. K-type and virulence factors of *Klebsiella* spp. were determined using Kleborate (v0.3.0, available at: <https://github.com/katholt/Kleborate>). Tn4401 variation was detected by TETyper [30].

2.4. Core genome phylogenies

Assembled genomes were annotated using Prokka (v1.12) before extracting core genome by Roary (v3.12.0), representing genes in ≥95% isolates of the same species [31,32]. It was not suitable to remove recombination regions before constructing a phylogenetic tree due to high diversity. A total of 2641 and 2268 core genes from *E. coli* and *K. pneumoniae*, respectively, were aligned to generate maximum likelihood trees based on general time reversible model with 1000 bootstrap replicates by IQ-TREE [33]. Dendrograms of CPE from different carbapenemase groups were built with and without accessory genomes by maximum likelihood estimation using default settings in Roary. Trees and dendrograms were visualized using ggtree and iTOL [34,35].

2.5. Plasmid mapping and long-read sequencing using MinION

Carbapenemase-encoding contig based on Illumina sequencing was extracted and online BLAST was performed for each CPE. The top 50 hits were downloaded and treated as reference plasmids. Short reads were then mapped to these hits using Bowtie 2 (v2.3.4.1) [36]. SAMtools (v1.9) was used to acquire the depth of each position and depth larger than 2 was used to calculate the coverage [37]. The plasmid with the highest coverage and coverage greater than 95% were treated as the best reference scaffold. CPE that shared the same reference plasmid were clustered to one group. Comparative views of plasmid sequences were generated with Easyfig (v2.2.2) [38]. Twelve representative strains from each group or strains with no best matches were selected for long-read sequencing. MinION library preparation was performed according to manufacturer protocol using ligation sequencing kit (SQK-LSK 109) and native barcoding expansion (EXP-NBD104) (Oxford Nanopore Technologies, UK). Libraries were poured into flow cell (R9.4.1) and run for 72 h. Long-read sequences were assembled with short-read sequences previously generated using Unicycler [16]. Isolates were assumed to acquire the backbone of the plasmid if their short-read sequences mapped $\geq 95\%$ of the reference plasmid.

2.6. Data availability

Raw reads of 169 isolates, hybrid assemblies and plasmid sequences were deposited into NCBI under the BioProject PRJNA595034.

3. Results

3.1. CPE distribution and carbapenemase groups

Among the 567 CPE isolates, 520 (91.7%) were isolated from CPE screening samples. The remaining 47 isolates were from urine (23), wound swab (5), respiratory samples (5), blood (2), and body fluid, tissue, and others (12).

The number of CPE isolated increased annually from a single isolate in 2011 to 238 isolates in 2017 (Figure 1A). The incidence rate of CPE steadily increased from 0.05 to 9.6 per 100 000 inpatient days for the three clusters of hospitals during this time (Figure S1). Only five groups of carbapenemase genes were detected over the 7-year period. Strains predominantly possessed a single type of carbapenemase gene ($n=553$), with *bla*_{NDM} ($n=298$, 52.6%) the most common, followed by *bla*_{IMP} ($n=169$, 30%), *bla*_{KPC} ($n=48$, 8.5%), *bla*_{OXA-48-like} ($n=27$, 4.8%), and *bla*_{IMI} ($n=11$, 1.9%). Fourteen isolates co-harbored carbapenemase genes, of which 12 co-harbored the *bla*_{NDM} and *bla*_{OXA-48-like} genes. The most common species in which these genes were harbored were *E. coli* (46.4%), *Klebsiella* spp. (38.3%), *Enterobacter* spp. (10.1%) and *Citrobacter* spp. (4.1%) (Figure 1B). All *bla*_{KPC} genes were present in *Klebsiella* spp., except one *E. coli*. *bla*_{IMI} was only present in *Enterobacter* spp. Strains from patients with no history of hospitalization overseas were defined as locally-acquired (67.4% of isolates). The proportion of CPE from patients with documented hospitalization overseas (Figure 1C) decreased even in 2014, but continued to reduce over time, contributing to less than 25% of cases with the expansion of screening criteria in 2016. This indicates the transition of increasing local/nosocomial acquisition from 2014 and 2015.

3.2. Genomic characterization of CPE

WGS was performed on 169 CPE strains of which 78 were *E. coli*, 66 were *Klebsiella* spp. (50 *K. pneumoniae*, 6 *K. oxytoca*, 7 *K. quasipneumoniae*, 2 *K. variicola*, 1 *Klebsiella quasivari-*

cola), 16 were *Enterobacter* spp., 6 were *Citrobacter* spp. and 3 were other species (1 *Serratia marcescens*, 1 *Raoultella planticola* and 1 *Leclercia adecarboxylata*). The most common alleles of each group of carbapenemase genes were *bla*_{NDM} ($n=100$): *bla*_{NDM-5} (51%), *bla*_{NDM-1} (42%), others (7%, *bla*_{NDM-4,7,9}); *bla*_{IMP} ($n=39$): *bla*_{IMP-4} (94.9%), *bla*_{IMP-26} (5.1%); *bla*_{KPC} ($n=22$): *bla*_{KPC-2} (100%); *bla*_{OXA-48-like} ($n=22$): *bla*_{OXA-181} (77.3%), *bla*_{OXA-232} (18.2%), *bla*_{OXA-48} (4.5%). Sequence types (ST) among the strains were diverse (Table S1). In *E. coli*, 32 STs were identified, of which ST131 ($n=5$) and ST542 ($n=5$) were identified carrying *bla*_{IMP} and ST405 ($n=6$) and ST48 ($n=4$) carrying *bla*_{NDM} were frequently detected. Twenty-five different STs were found in *K. pneumoniae*. Clonal Complex 258 (ST11, ST258) were common (90.0%) carrying *bla*_{KPC}, and ST38 were found in 4 isolates carrying *bla*_{NDM} plus *bla*_{OXA-48-like} genes.

3.3. Detection of multiple antimicrobial-resistant genes in CPE

CPE tend to carry multiple antimicrobial-resistant genes (ARGs), causing multidrug resistance (≥ 3 resistance classes). Isolates possessing *bla*_{NDM} plus *bla*_{OXA-48-like} genes carry, at most, 28 ARGs with a median of 10 resistance classes, which was significantly higher than other isolates ($P=0.004$, Welch's t-test) (Figure 2). Furthermore, 86 of 169 isolates (50.9%) had one or more extended spectrum beta-lactamases (ESBL) genes, with *bla*_{CTX-M-15}, *bla*_{SHV-12}, *bla*_{CTX-M-14}, *bla*_{CTX-M-55}, and *bla*_{OXA-10} identified in 25, 21, 12, 12, and 2 isolates, respectively. Nine CPE isolates (5.3%) were positive for colistin resistance *mcr*-like genes, including three variants, *mcr*-1, -3, and -9 (Table S2).

3.4. Diversity in serotypes and virulence genes of Klebsiella spp

The most prevalent capsular (K) loci in CP-K, *pneumoniae* were KL15 ($n=8$, including ST38, ST16, ST11, and ST585), KL64 ($n=8$, including ST11, ST14, ST147, and ST2497) and KL107 ($n=7$, all belonging to ST258). Of concern, one isolate belonging to serotype K1 with ST23 was identified carrying *bla*_{NDM-5}, which was the most prevalent hypervirulent type from liver abscess patients and stool samples in Hong Kong [39]. Seven types of yersiniabactin loci (*ybt*) were detected in 31.8% ($n=21$) of *Klebsiella* spp., with the combination of *ybt*-9 and mobile genetic elements ICEKp3 the most common ($n=8$). Other virulence genes were rare in the current collection, including three *iuc*-1 and one *rmpA2* positive strains.

3.5. Phylogeny of CPE

Core genome phylogenetic tree resolved IMP-positive *E. coli* into two distinct lineages (Figure 3). Isolates from lineage 1 (ST131) were local strains circulating in hospital cluster C, co-producing *bla*_{CTX-M-14} gene and two types of virulence factors (adhesin and siderophore). In contrast, lineage 2 (ST542) included strains from hospital cluster A and carried more AMR genes, such as *bla*_{CTX-M-55}, phenicol resistance gene *catA2* and tetracycline resistance gene *tet(B)*, but fewer virulence genes than lineage 1. On the contrary, the STs and the AMR composition were diverse in NDM-producing strains (Figure 3). Notably, the lineage of ST405 *E. coli* was associated with NDM-producing strains from patients with history of hospitalization in other Asian countries.

Two types of NDM plus OXA-48-like combination were identified for *K. pneumoniae*. In 2014, local strains producing NDM-1 plus OXA-181 in hospital cluster B were grouped in lineage 1, which belongs to ST38/KL15 (Figure 4). These isolates were extensively drug resistant, but no virulence factors were detected. However, the pattern changed to NDM-1 plus OXA-232 afterwards, carried by imported strains of diverse STs.

Two closely related lineages of KPC-positive isolates were found in hospital cluster A (Figure 4). Although they all belonged to

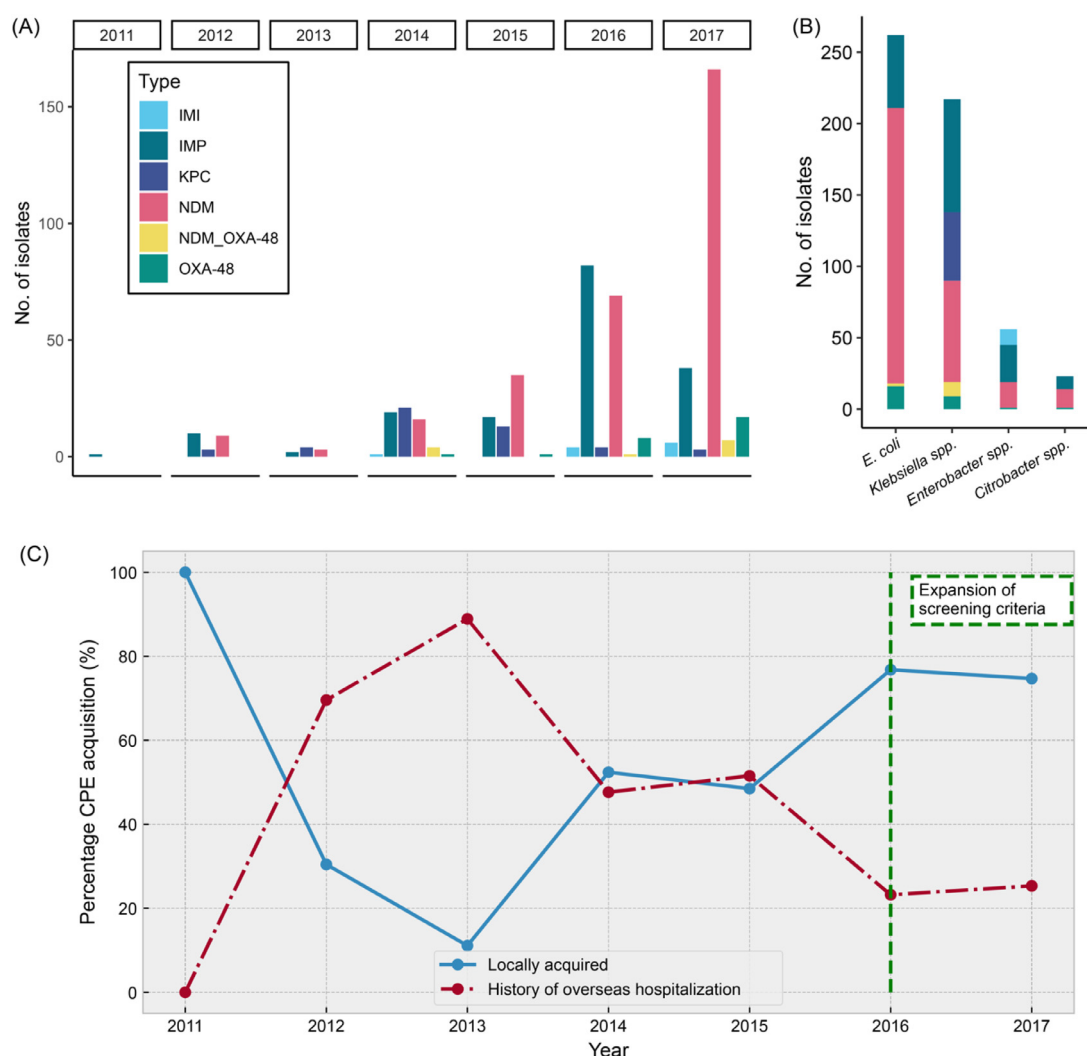


Figure 1. Distribution of 567 CPE isolates containing various carbapenemases from 2011 to 2017 is shown in (A). Major species are shown in (B). The proportion of CPE from patients with documented hospitalization overseas and locally-acquired during the same period is shown in (C). The CPE screening criteria have expanded since 2016 (refer to Methods).

Clonal Complex 258, the accessory genome profiles were different after screening for AMR genes and VFs. Lineage 2 contained ST11 isolates of serotype K64 or K47, carrying yersiniabactin *ybt-9* on ICEKp3, whereas, ST258/K107 strains in lineage 3 shared similar AMR patterns, resistant to more than six classes of antibiotics.

3.6. Plasmid analysis of CPE

The location of carbapenemase genes was determined in 92% of the sequenced CPE through short-read sequences mapping to plasmids that we have successfully circularized through hybrid assembly (12 strains as underlined in Figures S2 and S3) and from on-line reference sequences. Three types of IncX3 plasmids of different sizes (46 kb, 53 kb, 54 kb) were common (72.7%) in *bla*_{NDM}-carrying isolates and were distinguished by an extra ESBL gene (*bla*_{SHV-12}) or an insertion sequence (*ISAbA125*). However, *bla*_{NDM-1} of *bla*_{NDM} plus *bla*_{OXA-48-like} strains in *K. pneumoniae* were encoded on an IncFIB/IncHI1B plasmid (267 kb). Notably, this plasmid carried six extra AMR genes related to phenicol (*catA*, *catB*), quinolone (*qnrB*), aminoglycoside (*aac(6')-Ib-cr*) and cephalosporin (*bla*_{CTX-M-15}, *bla*_{OXA-1}) resistance. Other types of plasmids and their mapping coverages are shown in Figure S2A.

The IncX3 plasmids also harbored *bla*_{OXA-48-like} among *E. coli* and *Klebsiella* spp. When the plasmid harboring *bla*_{OXA-48-like} (CP034284) was aligned to that of *bla*_{NDM} (CP050157), there was 100% similarity at a coverage of 87.9%, indicating a closely related IncX3 backbone (Figure S2C). Another common plasmid found in *bla*_{NDM} plus *bla*_{OXA-48-like} strains was a short plasmid ColKP3 (6 kb) (Figure S2B).

For strains carrying *bla*_{IMP}, IncN-ST7 plasmids were detected across the 7 years of the study (Figure S3A). Genome alignment of three types of IncN plasmids demonstrated different insertion sequences downstream of *qnrS1* (Figure S3C). Also, *bla*_{IMP-26} was located on the 53 kb IncN-ST7 plasmid with transposon Tn2. The isolate that did not harbor IncN plasmid was hybrid assembled to a circular plasmid (CP050155) that has not been reported before, of which the *bla*_{IMP}-bearing region was mobilized to a 131 kb IncFII plasmid and flanked by two IS26 elements. Other ARGs were: *sul1*, *sul2*, *bla*_{DHA-1}, *qnrB*, *aph(3'')-Ib* and *aph(6)-Id*. One CPE strain mapped to IncN and IncFII plasmids indicated the presence of both plasmids in this isolate, thus the location of our carbapenemase gene cannot be inferred due to the limitation of short-read sequencing.

Eleven (50%) *bla*_{KPC} genes were located in transposon Tn4401a (10 kb), mainly in *K. pneumoniae* isolates belonging to ST258. Four

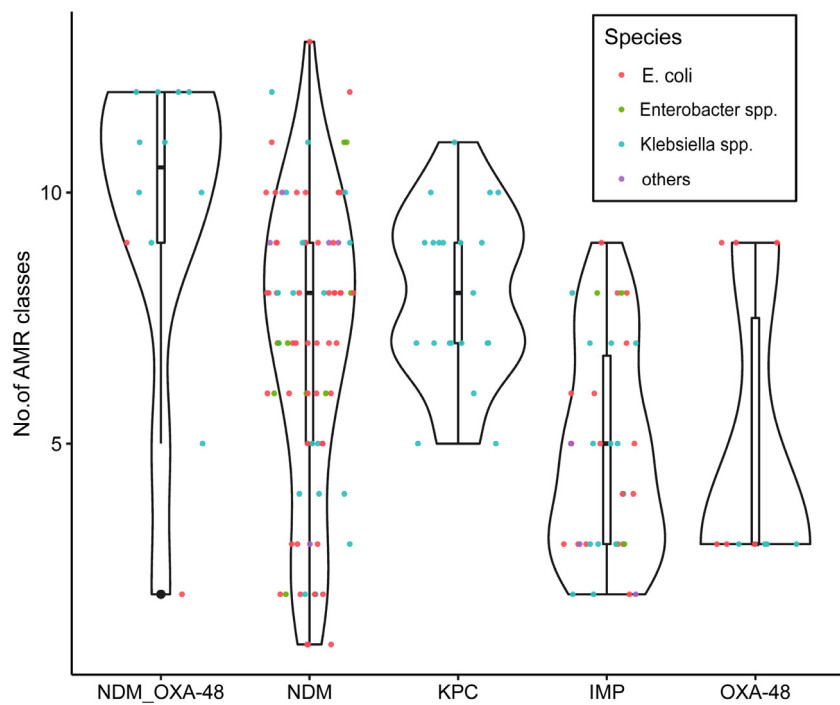


Figure 2. Number of antimicrobial resistance classes among different CPE strains. Species are represented by dots of specified colour.

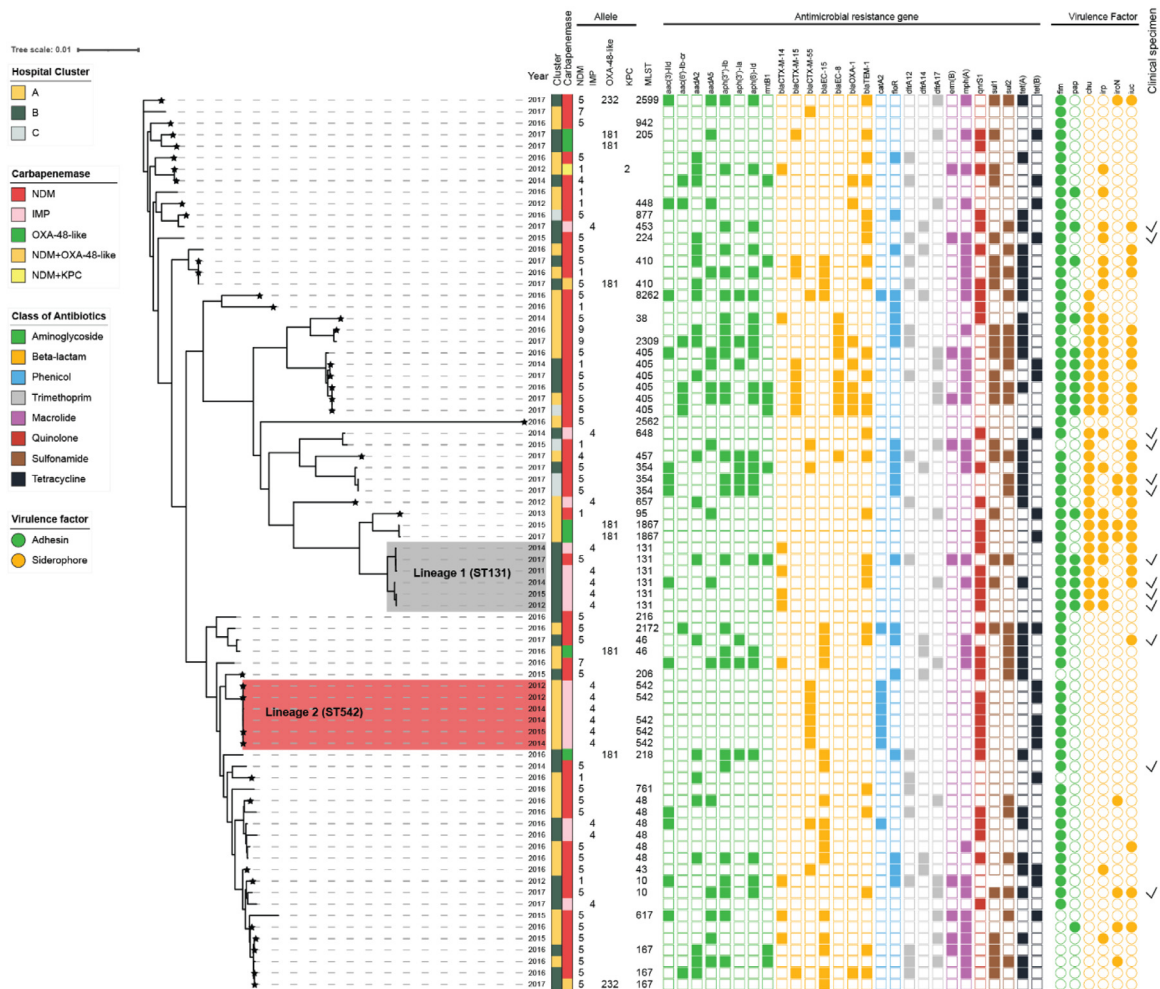
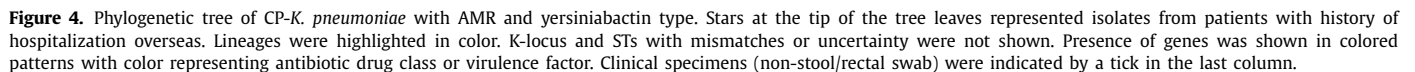


Figure 3. Phylogenetic tree of CP-*E. coli* with AMR and virulence profiles. Stars at the tip of the tree leaves represented isolates from patients with history of hospitalization overseas. Lineages were highlighted in color. STs with mismatches or uncertainty were not shown. Presence of genes was shown in colored patterns with color representing antibiotic drug class or virulence factor. Clinical specimens (non-stool/rectal swabs) were indicated by a tick in the last column.



Two distinct lineages (ST11, ST258) of *K. pneumoniae* in KPC-positive strains were observed in Hong Kong. ST11 *K. pneumoniae*-producing KPC is the predominant CPE in mainland China, of which the *bla*_{KPC} gene commonly transmitted through transposon Tn1721 [8]. However, only a few of the ST11 isolates carry Tn1721 in Hong Kong, and all others carried a truncated version (*ISKpn6-KPC-ISKpn8*), which has been previously reported in China [7]. The

first KPC-positive case (ST258 *K. pneumoniae*) reported in Hong Kong was a patient with travel history and medical follow-up in the United States in 2006 [43]. In the current study, besides ST11 *K. pneumoniae*, ST258 *K. pneumoniae* with transposon *Tn4401a* is also a vital cluster indicating multiple sources related to the transmission of KPCs among the CPE.

This study has several limitations. CPE isolates from only three of seven hospital clusters were included, thus the complete picture of the Hong Kong healthcare setting might be more complex than described. Prevalence may have been underestimated before the strengthening of screening policy in 2016. Although several factors were considered when selecting the most representative strains for WGS from the current cohort, WGS of all the available CPE isolates would provide a more comprehensive view. Confirmation of antimicrobial resistance by a phenotypic method was not performed in the current study. The ARGs were present in the genomes, thus the expression of the ARGs remained speculative.

In conclusion, this study described the rising trends of CPE isolated in hospitals in Hong Kong and the emergence of endemicity of CPE carriage from the local population, at least from hospitalized patients. The data summarized the prevalent groups of carbapenemases among Enterobacteriaceae in our hospitals in 2011–2017, as well as the main lineages and mobile genetic elements responsible for transmission and horizontal transfer of these resistance genes.

Declarations

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Ethical Approval: Approval was obtained from the respective Hospital Cluster Research ethics committee (KC/KE-17-0203/ER-4) and CUHK-NTEC CREC (ref no: 2017.152).

Author Contributions

MI, VL, KSCF, RWL, and DNT conceived and designed the study. CZ, CL, NWSL, KTW, NB, and YY performed the experimental work. RN, KYC, CKCL, MH, RWL, and KSCF contributed to the isolates and clinical data retrieval, and CZ, MI, and CL to the data analyses. CZ created the first draft of the manuscript and MI supervised the project. All authors contributed to the critical input and endorsed the final version of the manuscript.

Declaration of Competing Interest

None.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijantimicag.2021.106430](https://doi.org/10.1016/j.ijantimicag.2021.106430).

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